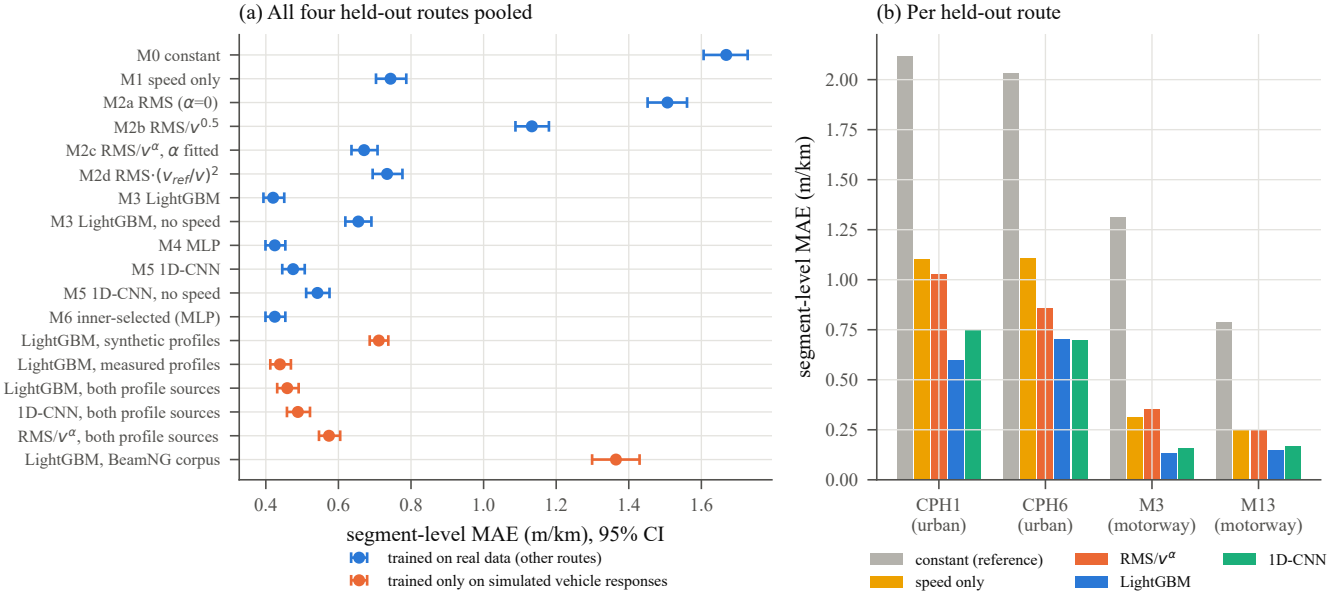
(a) All four held-out routes pooled
M0 constant
M1 speed only
M2a RMS (α=0)
M2b RMS/v^0.5
M2c RMS/v^α, α fitted
M2d RMS·(v_ref/v)^2
M3 LightGBM
M3 LightGBM, no speed
M4 MLP
M5 1D-CNN
M5 1D-CNN, no speed
M6 inner-selected (MLP)
LightGBM, synthetic profiles
LightGBM, measured profiles
LightGBM, both profile sources
1D-CNN, both profile sources
RMS/v^α, both profile sources
LightGBM, BeamNG corpus
0.4
0.6
0.8
1.0
1.2
1.4
1.6
segment-level MAE (m/km), 95% CI
trained on real data (other routes)
trained only on simulated vehicle responses
(b) Per held-out route
segment-level MAE (m/km)
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
2.00
CPH1 (urban)
CPH6 (urban)
M3 (motorway)
M13 (motorway)
constant (reference)
speed only
RMS/v^α
LightGBM
1D-CNN